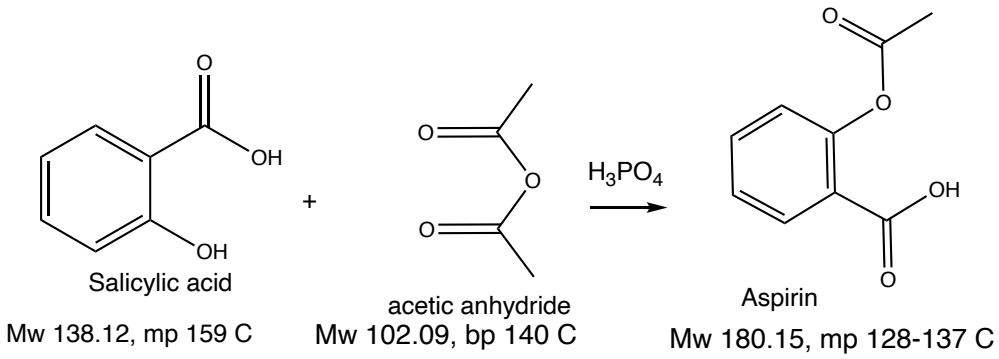
Synthesis of Acetylsalicylic Acid (Aspirin)

(Ref: Williamson, K.L and Masters, K.M., *Macroscale and Microscale Organic Experiments,* 7th Edition, Cengage, p. 534)



Salicylic acid
Mw 138.12, mp 159 C

acetic anhydride
Mw 102.09, bp 140 C

Aspirin
Mw 180.15, mp 128-137 C

Microscale Procedure

1. Take 138 mg of salicylic acid.
2. Add salicylic acid to a reaction tube. Add a boiling chip, and one small drop of 85% phosphoric acid followed by 0.3 mL (5 drops) of acetic anhydride (the acetic anhydride serves to wash the reactants to the bottom of the tube).
3. Mix the reactants thoroughly.
4. Heat the reaction tube on a hot water bath or in a beaker for 5 min. Temperature of water bath should be around 90°C.
5. Cautiously add 0.2 mL (3 drops) of water to the reaction mixture to decompose excess acetic anhydride. This will be an exothermic reaction.
6. When the reaction is over, add an additional 0.3 mL (5 drops) of water and allow the tube to cool slowly to room temperature.
7. If crystallization of the product does not occur during the cooling process, add a seed crystal or scratch the inside of the tube with a glass stirring rod.
8. Cool the tube in ice until crystallization is complete (at 10 least min) and then remove the solvent with a Pasteur pipette.
9. If the crystals are too fine for this procedure, collect the product by vacuum filtration on a Hirsch funnel.
10. Complete the transfer of the product to the funnel using a very small quantity of ice water. In either case, turn the product out onto a piece of filter paper and squeeze the crystals between sheets of filter paper to absorb excess water.
11. Allow the product to air-dry thoroughly before determining the weight and calculating the percent yield.
12. Determine the melting point and IR spectrum.

Aspirin is hydrolyzed by boiling water, but the reaction is not rapid; if desired, the product may be recrystallized from a small quantity of very hot water.

Cleaning Up. Dilute the filtrate with water and flush it down the drain.